

Zirong Chen

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EDUCATION

Vanderbilt University , Nashville, TN	Sept 2021 - May 2026
• <i>Doctor of Philosophy (Ph.D.)</i> in Computer Science; Advised by Dr. Meiyi Ma • <i>Research Interest</i>: Artificial Intelligence, Natural Language Processing, Large Language Models • <i>Dissertation Thesis</i>: Towards Trustworthy NLP-powered Intelligent Systems for Public Safety	
Georgetown University , Washington, DC	Sept 2019 - May 2021
• <i>Master of Science (M.S.)</i> in Computer Science • <i>Core Courses</i>: Deep Learning and Neural Nets, Empirical Methods in NLP, Deep Reinforcement Learning	
Jinling Institute of Technology , Nanjing, China	Sept 2015 - May 2019
• <i>Bachelor in Engineering (B.E.)</i> in Computer Science and Technology • <i>Thesis</i>: Security-Driven Big Data Applications in Smart Cities (with Nanjing Bureau of Public Security)	

SELECTED AWARDS AND HONORS

IJCAI Student Scholarship Awardee	June 2025
OpenAI Research Access Program Awardee (\$10,000)	Nov 2024
AAAI Student Scholarship Awardee	Feb 2024
Vanderbilt Student Travel Grant Awardee (x3)	2023; 2024; 2025

SELECTED PUBLICATIONS

IJCAI-2025 Zirong Chen, Ziyang An, Jennifer Reynolds, Kristin Mullen, Stephen Martini, Meiyi Ma. <i>LogiDebrief: A Signal-Temporal Logic Based Automated Debriefing Approach with Large Language Models Integration</i> . Accepted at IJCAI-2025 (13%), Montreal, Canada.
AAAI-2025 Zirong Chen, Elizabeth Chason, Noah Mladenovski, Erin Wilson, Kristin Mullen, Stephen Martini, Meiyi Ma. <i>Sim911: Towards Effective and Equitable 9-1-1 Dispatcher Training with an LLM-Enabled Simulation</i> . Published (18.9%) and Selected Oral Spotlight Presentation (4.9%) at AAAI-2025, Philadelphia, PA.
AAAI-2024 Zirong Chen, Xutong Sun, Yuanhe Li, Meiyi Ma. <i>Auto311: A Confidence-Guided Automated System for Non-emergency Calls</i> . Published (20.1%) and Selected Oral Spotlight Presentation (6%) at AAAI-2024, Vancouver, Canada.
Pervasive and Mobile Computing 2023 Zirong Chen, Isaac Li, Haoxiang Zhang, Sarah Preum, John A Stankovic, Meiyi Ma. <i>CitySpec with shield: A secure intelligent assistant for requirement formalization</i> . Invited journal extension at Elsevier's Pervasive and Mobile Computing (2023)
IEEE-SmartComp-2022 Zirong Chen, Isaac Li, Haoxiang Zhang, Sarah Preum, John A Stankovic, Meiyi Ma. <i>CitySpec: An intelligent assistant system for requirement specification in smart cities</i> . Published and Best Paper Finalist at IEEE SmartComp 2022, Helsinki, Finland.
AAMAS-2025 Ziyang An, Xia Wang, Hendrick Baier, Zirong Chen , Abhishek Dubey, Taylor T Johnson, Jonathan Sprinkle, Ayan Mukhopadhyay, Meiyi Ma. <i>Combining LLMs with a Logic-Based Framework to Explain MCTS</i> . Published (39.9%) at AAMAS-2025, Detroit, MI.

RESEARCH PROJECTS

Angie: AI-Driven Training System for 911 Call-takers in Emergency Responses	Dec 2023 - Present
<i>Integrating AI with real-world applications for a more efficient and effective emergency response training solution</i>	
• Led AI research project with Metro Nashville Emergency Communications, IT Services, and Government. Secured research grants from <i>OpenAI</i>, <i>Google</i>, <i>NSF(ReDDDoT)</i>, and <i>Vanderbilt</i>. • Served as major contributor and AI tech vendor, developing pioneering emergency response solution with sub-systems: <i>LogiDebrief</i>, <i>Sim911</i>, and <i>Auto311</i>. Deployed across 400+ workstations with 200+ active users,	

serving Metro Nashville for 14+ months.

LogiDebrief: An STL-based Automated 911 Call Debriefing System with LLMs

Nov 2024 - Present

Disentangle overly complicated prompts into logic-enforced specifications with modularized use of LLMs

- Developed signal temporal logic (STL) framework integrated with LLMs to rigorously analyze and debrief historical 911/311 calls, maintaining accuracy across extended context windows. Through extensive real-world testing, LogiDebrief achieves over 95% accuracy without performance degradation on long contexts.
- Successfully deployed at local emergency response center for both daily operations and call-taker training programs. The system has automated analysis of over 1,701 emergency calls, resulting in 311+ hours of operational time savings.

Sim911: An LLMs-based Simulation System for Call-taker Training

Mar 2024 - Present

An effective and reliable workflow for LLMs role-play simulation under emergency responses

- Developed context-aware generation and validation framework using CoT, RAG, and few-shot prompting with LLM agents to ensure realistic and inclusive simulations (addressing underserved groups). Framework incorporates automatic correction based on human feedback for continuous improvement.
- In real-world training programs, 90%+ participants rated Sim911 as comparable or superior to human-led simulations. Framework outperforms baseline by 62% in authenticity benchmarks.
- Deployed in local emergency response center's call-taker training programs. Generated 600+ simulated calls with 40+ hours of active engagement, saving 80+ hours of instructor time.

Auto311: A Confidence-Guided Automated System for Non-emergency Calls

Jan 2023 - Present

Integrating AI with real-world applications for an automated call handling system

- Collaborated with Metro Nashville Department of Emergency Communications, Metro Nashville Information Technology Services, and Metro Nashville Government.
- Designed and developed an automatic call handling system for non-emergency calls based on confidence measurement with deep neural networks. Outperformed naïve LLMs calls with 94.49% acc under emulation. Implemented baseline models like Transformers.

INTERNSHIP EXPERIENCES

Research Fellow, Epiq AI Labs (formerly LAER AI) – New York, NY

May 2025 – Aug 2025

- Enhanced AI Discovery AssistantTM with **DeepResearch** capabilities through agentic workflow design, optimized for eDiscovery, document review, and complex legal scenarios.
- Created **BREEW** (Bayesian Reasoning via Entity-Evidence Weaving) benchmark that leverages concept decomposition and entity-level evidence surfacing to ensure logical coherence and rigorous concept validation.
- Built **LINT** (Learning via Interactive Tutoring) testing environment featuring simulated user agents with restorable states. The system evaluates agent performance through controlled conversational interactions using probabilistic finite state machines and configurable personas.

Data Scientist^(NDA), National Basketball Association (NBA) – New York, NY

May - Aug 2020, 2021

- Analyzed user feedback using neural networks (PyTorch, Keras) and NLP tools (NLTK, spaCy) to develop data-driven fan engagement strategies.
- Enhanced marketing campaign effectiveness by applying statistical ML models (scikit-learn) to optimize user experience and advertisement targeting.
- Developed real-time data analysis pipelines with sequential neural networks (RNNs, Time2Vec) to predict streaming bandwidth requirements, optimizing infrastructure costs and user experience.
- Architected scalable and reusable neural network frameworks for cross-project deployment, improving development efficiency and model adaptability.

SELECTED TALKS AND SERVICES

Invited Educational AI Demo at LIVE Learning Innovation Incubator

Mar 2024, 2025

Guest lectures at Vanderbilt (AI in CPS): “Large Language Models”

Feb 2023, 2024; Oct 2025

Guest lectures at Vanderbilt (AI): “Natural Language Processing”

Oct 2022, 2023, 2024

Conference/Journal Program Committee (e.g., ICLR, AAAI, ACM Trans, etc.)